

The stomach as a bioreactor: dietary lipid peroxidation in the gastric fluid and the effects of plant-derived antioxidants.

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Atherosclerosis may result partly from processes that occur following food consumption and that involve oxidized lipids in chylomicrons. We investigated reactions that could occur in the acidic pH of the stomach and accelerate the generation of lipid hydroperoxides and co-oxidation of dietary constituents. The ability of dietary polyphenols to invert catalysis from pro-oxidation to antioxidation was examined. The acidic pH of gastric fluid amplified lipid peroxidation catalyzed by metmyoglobin or iron ions. Metmyoglobin catalyzed peroxidation of edible oil, resulting in 8-fold increase of hydroperoxide concentration. The incubation of heated muscle tissue in simulated gastric fluid for 2 h enhanced hydroperoxides accumulation by 6-fold to 1200 microM. In the presence of catechin or red wine polyphenols, metmyoglobin catalyzed the breakdown of hydroperoxides to zero, totally preventing lipid peroxidation and beta-carotene cooxidation. We suggest that human gastric fluid may be an excellent medium for enhancing the oxidation of lipids and other dietary constituents. The results indicate the potentially harmful effects of oxidized fats intake in the presence of endogenous catalysts found in foods, and the major benefit of including in the meal plant dietary antioxidants.